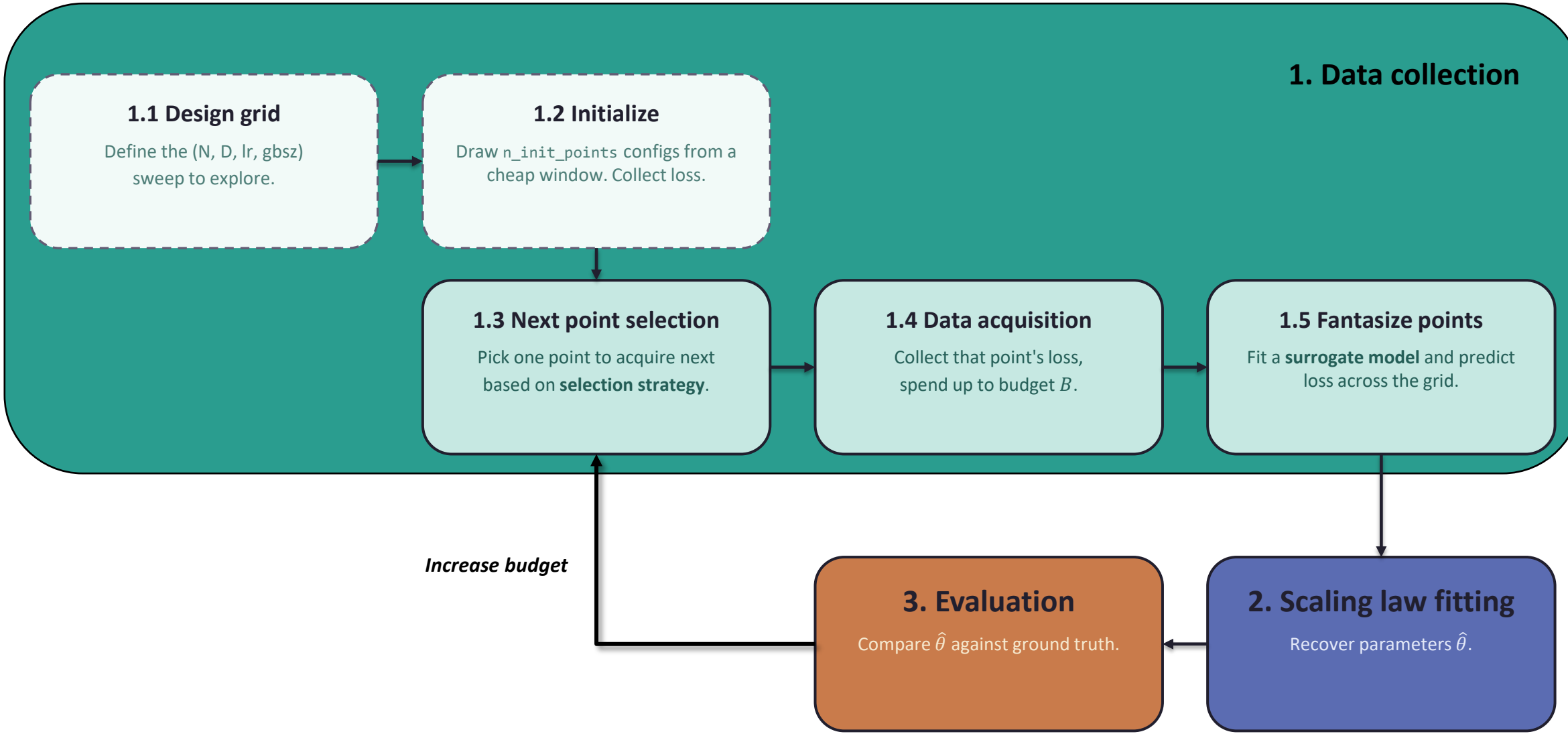

1. Data collection
1.1 Design grid
Define the (N, D, lr, gbsz) sweep to explore.
1.2 Initialize
Draw n_init_points configs from a cheap window. Collect loss.
1.3 Next point selection
Pick one point to acquire next based on selection strategy.
1.4 Data acquisition
Collect that point's loss, spend up to budget B.
1.5 Fantasize points
Fit a surrogate model and predict loss across the grid.
2. Scaling law fitting
Recover parameters $\hat{\theta}$.
3. Evaluation
Compare $\hat{\theta}$ against ground truth.
Increase budget